Sheaf Reconstruction: Cohomological Consistency for Multi-View Pointmap Fusion in Dynamic Scenes

Author One
Department / Institution
author1@institution.edu

Author Two
Department / Institution
author2@institution.edu

June 4, 2026

Abstract

Feed-forward pointmap regression has become the dominant route to dense multi-view reconstruction: a network maps each image pair to per-pixel 3D coordinates in a shared frame, so that depth, relative pose, and dense correspondence emerge together. Turning many such pairwise predictions into a single scene still requires a *fusion* step, handled today either by a per-scene global-alignment optimization or by a learned recurrent state. Neither makes explicit *where*, and to *what degree*, the pairwise predictions fail to admit a globally consistent reconstruction—a question that is a nuisance for rigid scenes but central for dynamic ones, where object motion makes a single consistent rigid geometry genuinely obstructed rather than merely hard to find. We recast multi-view pointmap fusion as the search for a *global section* of a cellular sheaf on the view graph, whose stalks carry per-view geometry and whose restriction maps carry the network's predicted relative transforms. In this language a globally consistent reconstruction is the kernel of the sheaf Laplacian, recoverable by a sparse, graph-native diffusion with a spectrally controlled rate, while the obstruction to consistency is measured by the first cohomology, whose mass localizes on exactly the moving and occluded regions. The formulation subsumes the correspondence graph as a strictly less expressive special case, yields a principled alternative to the opaque global-alignment optimizer, and produces an unsupervised dynamic-region detector as a by-product. This document develops the formulation and the preliminaries it rests on.

1 Introduction

Pointmap regression has reshaped multi-view 3D reconstruction. In place of the classical pipeline—detect correspondences, estimate poses, triangulate, fuse—models in the DUSt3R family [Wang et al., 2024, Leroy et al., 2024] learn a single feed-forward map from an image pair (I^1, I^2) to two dense *pointmaps*, per-pixel 3D coordinates expressed in a common camera frame, so that depth, relative pose, and dense correspondence are read off a single prediction. For dynamic scenes, where object motion breaks the rigid-alignment assumption, MonST3R [Zhang et al., 2025] retrains on moving content and D^2USt3R [Han et al., 2026] introduces static–dynamic aligned pointmaps that warp static pixels by camera pose and dynamic pixels by optical flow, sharpening correspondence on moving objects.

These models are inherently *pairwise*. Recovering a scene from $N > 2$ views requires a separate *fusion* stage that reconciles the overlapping pairwise predictions into one globally consistent geometry. Two families address it. The first keeps a per-scene *global-alignment* optimization: the original DUSt3R formulation minimizes pairwise pointmap disagreement over per-view depths, poses, and intrinsics by gradient descent for a few hundred iterations [Wang et al., 2024], an objective inherited with minor variation by its dynamic successors. The second replaces optimization with a learned state: Spann3R carries an external spatial memory [Wang and Agapito, 2025], CUT3R maintains a persistent recurrent state that emits each new pointmap directly in a shared frame [Wang et al., 2025], and SLAM-oriented systems fuse submaps with loop closures [Murai et al., 2025].

Across both families, global consistency is enforced *implicitly*. The optimizer drives pairwise residuals toward zero; the learned state absorbs multi-view agreement into its weights. Neither exposes the object that actually controls whether fusion can succeed: a *principled, computable account of the obstruction* to gluing the local predictions into one scene. For rigid scenes this absence is tolerable, because a consistent global geometry exists and the only question is how fast one finds it. For dynamic scenes it is the crux. A moving object photographed at two times has *no* single rigid placement consistent with both views; the pairwise predictions are locally correct yet globally incompatible, and the incompatibility is not noise to be averaged away but structure to be measured. Current fusion either fights this obstruction (rigid alignment, which smears the error into neighboring static geometry) or hides it (a learned state, which offers no handle on where consistency genuinely failed).

Our viewpoint. The pairwise predictions carry exact algebraic structure that names this obstruction. Index the views by the vertices of a *view graph* $G = (V, E)$, with an edge for each pair the network has related. Each view supplies a local description of the scene's geometry; each edge supplies a predicted transform relating the two descriptions on their overlap. A globally consistent reconstruction is precisely an assignment of a geometry to every view that agrees, edge by edge, under these transforms—in the language of applied topology, a *global section* of a cellular sheaf [Curry, 2014, Ghrist, 2014] whose stalks are the per-view geometries and whose restriction maps are the predicted transforms. We make this identification precise and build the fusion problem on it. The sheaf is the canonical structure for integrating local observations into a global state [Robinson, 2017]; our contribution is to recognize that multi-view pointmap fusion is an instance, and that its dynamic-scene difficulties are exactly the sheaf's cohomology.

What the viewpoint buys. Three consequences follow, and they organize the paper. *(i) Consistency is a kernel.* A reconstruction is globally consistent if and only if it lies in the kernel of the *sheaf Laplacian* $\mathbf{L}_{\mathcal{F}}$, a sparse positive-semidefinite operator built from the restriction maps [Hansen and Ghrist, 2019]. Fusion becomes a graph-native diffusion—heat flow under $\mathbf{L}_{\mathcal{F}}$, a sequence of sparse linear solves—whose convergence rate is governed by the spectral gap of $\mathbf{L}_{\mathcal{F}}$, replacing the opaque gradient descent of global alignment with an operator one can analyze and precondition. *(ii) Inconsistency is cohomology, and it localizes.* When no global section exists, the failure is measured by the first cohomology $H^1(G; \mathcal{F})$, equivalently by the harmonic 1-cochains in the cokernel of the coboundary. In a dynamic scene this obstruction mass concentrates on the edges and regions where motion and occlusion make local predictions globally irreconcilable, yielding an *unsupervised* dynamic-region detector—reaching D^2USt3R's static/dynamic split from topology rather than from an optical-flow threshold. *(iii) The sheaf strictly refines the graph.* A bare correspondence graph records *that* two views are related; the sheaf records *how*, by carrying the transition maps as data, and its cohomology expresses global obstructions—scale drift, loop inconsistency, motion—that a flat union of pointmaps cannot represent. This answers, in the affirmative and precisely, whether a graph-structured world model is more expressive than a pointmap pile: not the graph, but the sheaf on it, is.

Relation to prior structure. The closest existing machinery is group synchronization: estimating absolute poses on graph vertices from relative measurements on edges [Singer, 2011], with certifiable solvers over $\mathrm{SE}(3)$ [Rosen et al., 2019] and, paired with a learned depth prior, over the similarity group [Yu and Yang, 2024]. Synchronization is the *linear, single-stalk* shadow of our formulation—the special case in which every stalk is one copy of the pose group and the only data are relative transforms—and it targets the same fusion stage. The sheaf view generalizes it in two ways that matter for dynamic scenes: stalks may carry dense per-view geometry rather than a single pose, and the cohomological obstruction is retained and localized rather than relaxed away. On the topology side, spectral sheaf theory [Hansen and Ghrist, 2019, 2021] and sheaf-based sensor fusion [Robinson, 2017] supply the operators we use; our contribution is their first instantiation, to our knowledge, on dense multi-view pointmap reconstruction, together with the dynamic-scene reading of H^1.

Contributions.

1. We recast multi-view pointmap fusion as recovering a global section of a cellular sheaf on the view graph, with per-view geometry as stalks and the network's predicted transforms as restriction maps (Section 2.3). The construction subsumes both DUSt3R-style global alignment and $\mathrm{SE}(3)$/$\mathrm{Sim}(3)$ synchronization as special cases.
2. We characterize globally consistent reconstructions as the kernel of the sheaf Laplacian and give a sparse diffusion solver with a convergence rate controlled by its spectral gap, providing a principled, analyzable replacement for the global-alignment optimizer.
3. We show that the first cohomology localizes the obstruction to consistency, and that in dynamic scenes its mass concentrates on moving and occluded regions, yielding an unsupervised dynamic-region detector that requires no flow estimator.

4. We analyze the formulation's two honest failure modes—the non-abelian curvature of pose-valued stalks, handled by a connection (tangent-space) linearization, and the coupling between the evolving reconstruction and the sheaf it induces—and validate the resulting method empirically.

2 Preliminaries

We fix notation for pointmaps and the fusion problem (Section 2.1), recall cellular sheaves and their spectral theory (Section 2.2), and assemble the two into the reconstruction sheaf that the method operates on (Section 2.3).

2.1 Pointmaps and the fusion problem

Let $I^1, \ldots, I^N$ be images of resolution $W \times H$. A *pointmap* $X^{n,m} \in \mathbb{R}^{W \times H \times 3}$ assigns to each pixel of image n its 3D location expressed in the camera frame of image m. For a pair (n, m) the network of the DUSt3R family [Wang et al., 2024, Zhang et al., 2025, Han et al., 2026] regresses $X^{n,m}$ and $X^{m,m}$ together with per-pixel confidences $C^{n,m}$, from which depth, the relative camera transform, and dense correspondence follow. Writing $\pi_K^{-1}(p, d) = d\, K^{-1}\tilde{p}$ for the back-projection of pixel p at depth d through intrinsics K ($\tilde{p}$ homogeneous), each view n carries a depth map D_n, a pose $g_n \in \mathrm{SE}(3)$, and intrinsics K_n, and contributes the world-frame points

$$\chi_n(p) \;=\; g_n \cdot \pi_{K_n}^{-1}\big(p,\, D_n(p)\big). \tag{1}$$

The view graph. Collect the views as the vertices of a *view graph* $G = (V, E)$, placing an edge $e = (i, j) \in E$ whenever the network has produced a pairwise prediction relating views i and j (in practice, pairs selected by a confidence- or overlap-based scene graph). Each edge carries the predicted relative geometry on the overlap of the two views: a relative transform g_e, a relative scale σ_e, and the correspondence field that identifies which pixels of i and j view the same scene point.

Global alignment. The standard fusion objective reconciles the pairwise predictions by minimizing a confidence-weighted pairwise disagreement,

$$\mathcal{L}_{\text{align}} \;=\; \sum_{e=(i,j)\in E}\; \sum_{n\in\{i,j\}}\; \sum_{p} \nu_n(p)\, \Big\| \chi_n(p) \;-\; g_e\Big(\sigma_e\, X_n^{(e)}(p)\Big) \Big\|, \tag{2}$$

over the per-view variables $\{D_n, g_n, K_n\}$ and per-edge variables $\{g_e, \sigma_e\}$, solved by gradient descent [Wang et al., 2024]. Two features of (2) motivate what follows. First, it couples the variables *only* through the shared correspondence encoded on edges—it is a sum of edge-local agreement terms—so its structure is that of a function on a graph, not on an unstructured point cloud. Second, it presumes a consistent rigid global geometry exists and seeks it by descent; it has no notion of, and returns no certificate for, the case where consistency is genuinely obstructed, which is the generic situation under object motion. The reformulation below keeps the edge-local structure of (2) but replaces "descend on the residual" with "identify the residual's consistent part and measure its obstruction."

2.2 Cellular sheaves and their Laplacian

We recall the minimal sheaf-theoretic vocabulary; see Curry [2014] and Hansen and Ghrist [2019] for full treatments.

Definition 1 (Cellular sheaf)**.** A *cellular sheaf* $\mathcal{F}$ of finite-dimensional real vector spaces on a graph $G = (V, E)$ assigns

- to each vertex $v \in V$ a *stalk* $\mathcal{F}(v)$, a vector space of data local to v;
- to each edge $e \in E$ a stalk $\mathcal{F}(e)$, the data shared on e;
- to each incident pair $v \trianglelefteq e$ a linear *restriction map* $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}(v) \to \mathcal{F}(e)$.

The 0-*cochains* $C^0(G; \mathcal{F}) = \bigoplus_{v\in V} \mathcal{F}(v)$ collect a choice of local data at every vertex; the 1-*cochains* $C^1(G; \mathcal{F}) = \bigoplus_{e\in E} \mathcal{F}(e)$ collect data on every edge. Fixing an orientation on each edge $e = (i, j)$, the *coboundary* $\delta : C^0 \to C^1$ compares the two endpoints through their restriction maps,

$$(\delta x)_e \;=\; \mathcal{F}_{j \trianglelefteq e}\, x_j \;-\; \mathcal{F}_{i \trianglelefteq e}\, x_i, \qquad e = (i, j). \tag{3}$$

Definition 2 (Global sections)**.** A 0-cochain $x \in C^0(G; \mathcal{F})$ is a *global section* if $\delta x = 0$, i.e. if the local data agree on every edge after restriction. The space of global sections is $\Gamma(G; \mathcal{F}) = \ker \delta = H^0(G; \mathcal{F})$.

Equipping every stalk with an inner product makes δ a map of inner-product spaces with adjoint $\delta^\top$. The *(degree-0) sheaf Laplacian* is

$$\mathbf{L}_{\mathcal{F}} \;=\; \delta^\top \delta \;:\; C^0(G;\mathcal{F}) \to C^0(G;\mathcal{F}), \tag{4}$$

a sparse, symmetric, positive-semidefinite operator whose (i, j) block is supported exactly on the edges of G, so that $\mathbf{L}_{\mathcal{F}}$ inherits the sparsity of the view graph. When every stalk is $\mathbb{R}$ and every restriction map is the identity, $\mathbf{L}_{\mathcal{F}}$ is the ordinary graph Laplacian; the sheaf Laplacian is its data-carrying refinement. We use two standard facts.

Lemma 1 (Sections are the kernel; obstruction is cohomology [Hansen and Ghrist, 2019])**.** *For a cellular sheaf $\mathcal{F}$ on a graph,*

$$\ker \mathbf{L}_{\mathcal{F}} \;=\; \ker \delta \;=\; \Gamma(G;\mathcal{F}) \;=\; H^0(G;\mathcal{F}),$$

and the first cohomology $H^1(G;\mathcal{F}) = C^1/\operatorname{im}\delta \cong \ker \delta^\top$ is isomorphic to the space of harmonic 1-cochains. *In particular $H^1 = 0$ if and only if every edge residual can be explained by a global section.*

Lemma 1 is the structural backbone of the method: recovering a consistent reconstruction means projecting onto $\ker \mathbf{L}_{\mathcal{F}}$, which the heat flow $\dot{x} = -\mathbf{L}_{\mathcal{F}}\, x$ achieves with a rate set by the smallest nonzero eigenvalue of $\mathbf{L}_{\mathcal{F}}$; and the residual that no global section can absorb is the harmonic representative of $H^1(G;\mathcal{F})$, a quantity supported on edges and therefore *localizable*.

2.3 The reconstruction sheaf

We now place the fusion problem of (2) on the sheaf of Section 2.2. Fix the view graph $G = (V, E)$ of Section 2.1.

Stalks. The stalk $\mathcal{F}(v)$ over a view holds that view's contribution to the world geometry. In the simplest *coordinate* model, $\mathcal{F}(v)$ is the space of world-frame point assignments χ_v of (1) for the pixels of view v; in the *pose* model used for analysis, $\mathcal{F}(v)$ is the tangent space at the current estimate of view v's frame and scale, i.e. a copy of the Lie algebra $\mathfrak{se}(3) \oplus \mathbb{R}$ (the generator of $\mathrm{Sim}(3)$). The edge stalk $\mathcal{F}(e)$ over $e = (i, j)$ is the space of geometry on the shared overlap of views i and j, into which both endpoints restrict.

Restriction maps. The restriction map $\mathcal{F}_{i \trianglelefteq e}$ takes view i's local geometry to its prediction on the overlap of e; concretely it applies the back-projection (1) and selects the pixels of i that the edge's correspondence field matches to j. The two restriction maps of an edge are built so that $(\delta x)_e = 0$ encodes *exactly* the per-edge agreement term inside (2): a global section is a set of per-view geometries whose overlaps coincide after the predicted transforms, which is the definition of a globally consistent reconstruction. Thus $\mathcal{L}_{\mathrm{align}}$ in (2) is, up to the confidence weights $\nu_n(p)$ folded into the stalk inner products, the Dirichlet energy $\frac{1}{2}\,\langle x,\, \mathbf{L}_{\mathcal{F}} x\rangle = \frac{1}{2}\,\|\delta x\|^2$ of the reconstruction sheaf.

Proposition 1 (Fusion as a kernel problem; informal)**.** *With the reconstruction sheaf above, the minimizers of the global alignment energy* (2) *are the 0-cochains nearest to* $\ker \mathbf{L}_{\mathcal{F}} = \Gamma(G;\mathcal{F})$ *in the stalk inner product. When a consistent rigid reconstruction exists it is a global section and the energy attains zero; when object motion or loop inconsistency obstructs consistency, the irreducible residual is the harmonic representative of $H^1(G;\mathcal{F})$.*

Proposition 1 reframes fusion from descent on a residual to projection onto a kernel, and it identifies the leftover—the part of the data that *cannot* be made consistent—with a cohomology class rather than with optimizer noise. The dynamic-scene method developed in the sequel reads this class off the harmonic 1-cochains and attributes its support to moving and occluded regions.

Remark 1 (Two honest subtleties). First, poses live in $\mathrm{Sim}(3)$, which is non-abelian, so the linear coboundary (3) is exact only on the tangent stalks of the pose model; on the group itself the correct object is a sheaf with a *connection*, whose curvature around a cycle is the synchronization obstruction [Singer, 2011, Rosen et al., 2019]. We work in the tangent linearization and re-linearize as the estimate updates, a sheaf-valued Gauss–Newton iteration. Second, the restriction maps depend on the correspondence field, which depends on the very geometry we solve for, so the sheaf co-evolves with its section; we treat the network's pairwise predictions as fixed during each solve and refresh them between outer iterations. Both subtleties are analyzed in Section [TODO: method].

3 A Proof-of-Concept Experiment

The viewpoint of Section 2.3 rests on one empirical claim that no prior theory supplies: that the harmonic representative of $H^1(G;\mathcal{F})$ concentrates on moving regions rather than smearing into the static surround or being swamped by loop-closure and drift modes (Contribution 3). Before committing to the non-abelian machinery of Remark 1, we test this claim in the cleanest setting that can falsify it: a controlled synthetic scene with a single known rigid moving object, on which we build the linearized reconstruction sheaf and compute the harmonic 1-cochain numerically. The

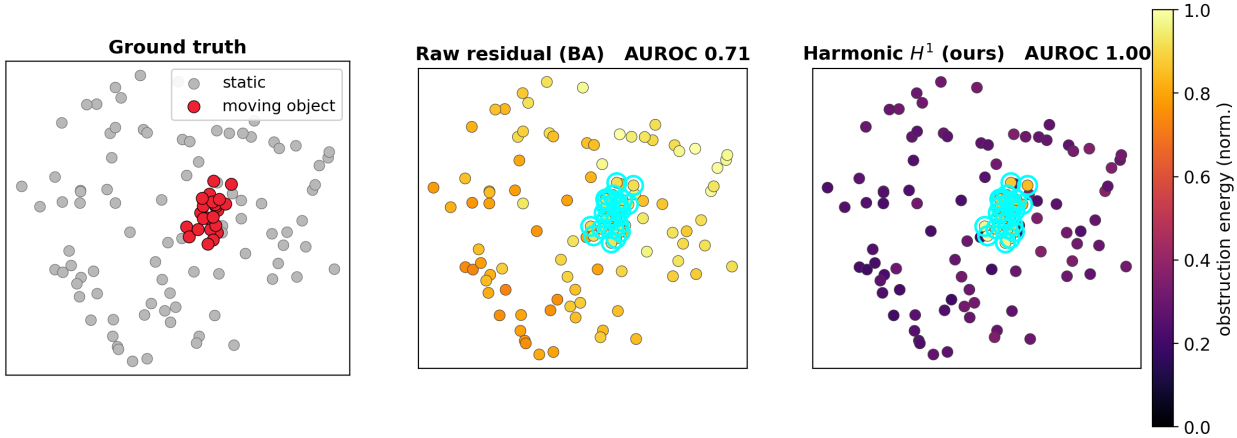

Figure 1: **The harmonic H^1 representative localizes on the moving object where the raw residual cannot.** Per-point energy on a scene with accumulating camera drift (0.30 rad/step), one projection; cyan rings mark the ground-truth moving object. *Left:* ground truth. *Middle:* the raw residual energy R_k that bundle adjustment minimizes—drift inflates static points too, so its brightest points are static and the object is not separable (AUROC 0.71). *Right:* the harmonic energy E_k, which removes everything a pose correction can explain and lights up exactly the moving object (AUROC 1.00). Energies are min–max normalized per panel.

experiment uses no learned network and no training; it isolates *the sheaf machinery* from the quality of any particular pairwise predictor, and runs in seconds on CPU.

3.1 Setup

We place $N = 14$ cameras on a partial orbit viewing a scene of 90 static background points and one rigid object of 24 points that translates and rotates between frames. The view graph G is the temporal chain augmented with a few long-range loop-closure edges, so that cycles—and hence a place for loop-inconsistency modes to live—are present. We instantiate the reconstruction sheaf in the pose model: each vertex stalk is the $\mathrm{Sim}(3)$ tangent $\mathfrak{se}(3) \oplus \mathbb{R}$, and the edge stalk over $e = (i, j)$ is decomposed *per co-visible point*, one copy of $\mathbb{R}^3$ per point, so that a 1-cochain carries a residual for each scene point on each edge and H^1 can localize spatially rather than merely per view. For point k on edge (i, j), each endpoint pushes its observation through the current pose estimate to a world point w_k^i, giving the base residual $r^0_{ij,k} = w_k^i - w_k^j \in \mathbb{R}^3$; the coboundary acts by the left-perturbation Jacobian $B_{i,k} = [\, I \;\; -[w_k^i]_\times \;\; w_k^i \,]$, and the sheaf Laplacian is $\mathbf{L}_{\mathcal{F}} = \delta^\top \delta$, which is precisely the bundle-adjustment information matrix. The harmonic representative is the part of the residual that no pose correction can remove, $h = r^0 + \delta\xi^\star$ with $\xi^\star = \arg\min_\xi \left\| r^0 + \delta\xi \right\|^2$, obtained by a sparse least-squares solve (projection onto $\ker \delta^\top$, cf. Lemma 1). We compare the per-point harmonic energy $E_k = \sum_{e \ni k} \|h_{e,k}\|^2$ against the per-point raw residual energy $R_k = \sum_{e \ni k} \left\| r^0_{e,k} \right\|^2$—the quantity bundle adjustment drives to zero—and score how well each separates the (known) moving points from the static ones by AUROC.

3.2 The obstruction localizes, and survives drift

Figure 1 shows the central result. With ground-truth poses both E_k and R_k trivially isolate the object; the interesting case is *accumulating odometry drift*, under which the static background also disagrees across views. The raw residual then conflates that disagreement with object motion and fails to localize (AUROC 0.71, and its brightest points are static), whereas the harmonic projection first subtracts off every pose-explainable disagreement and retains only the irreducible part, which sits squarely on the moving object (AUROC 1.00). This is exactly the distinction Proposition 1 draws: bundle adjustment treats the residual as noise to be minimized and discards the leftover, while the cohomological reading keeps it as a class and reads it as scene dynamics. Figure 2 quantifies the robustness across a sweep of drift magnitudes: the harmonic AUROC is pinned at 1.00 with zero variance while the raw AUROC decays from 1.00 toward chance, with a variance band that crosses below 0.5. The harmonic signal is, empirically, invariant to the camera drift that defeats the raw residual.

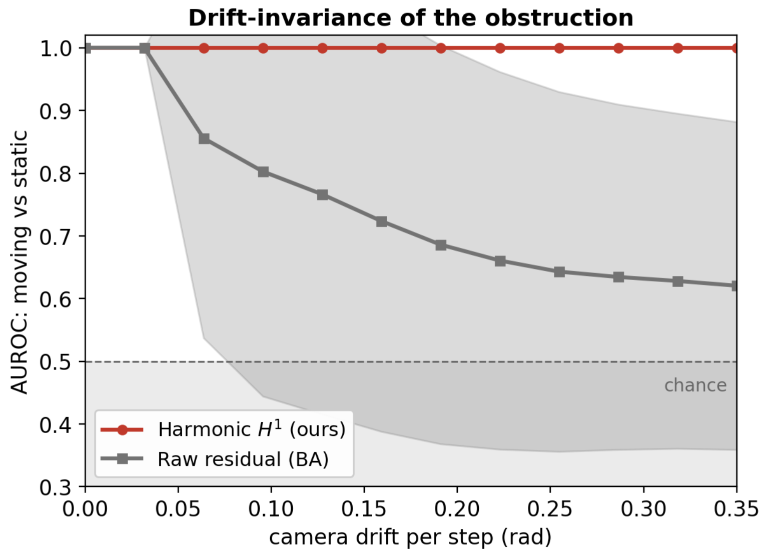

Figure 2: **The cohomological signal is invariant to camera drift; the raw residual is not.** Localization AUROC (moving vs. static) versus accumulating per-step camera drift, mean $\pm$ s.d. over 6 seeds. The harmonic energy stays at 1.00 because drift is, by construction, in the image of the coboundary and is projected away; the raw residual decays from 1.00 toward chance as drift grows. The two coincide only at zero drift—the regime the dynamic-scene problem is not about.

3.3 Honest boundaries

The experiment also makes three limitations concrete, each of which sharpens the theory to be built.

The global scale gauge must be anchored. The coordinate residual $w_k^i - w_k^j$ is not invariant under a global rescaling of the reconstruction, so the uniform log-scale direction of the $\mathrm{Sim}(3)$ stalks is a coboundary that shrinks the world toward a point and annihilates *every* residual, the obstruction included; left free, it drives the harmonic energy and the localization AUROC to zero. Anchoring the global scale (one linear constraint pinning $\sum_v$ of the per-view log-scales) removes this mode and restores the result. This is the metric-scale gauge of $\mathrm{Sim}(3)$ reconstruction surfacing in the spectral picture, and it should be stated alongside the non-abelian subtlety of Remark 1.

Localization presumes a static-dominant scene. The static background is what pins the remaining gauge: it is the data that fixes the per-view frames, leaving the moving object as the unexplained residual. As the static fraction falls, this anchoring weakens, and localization degrades sharply once moving points cease to be a minority—past the point where static and dynamic points are equinumerous, the projection can no longer tell which is the scene and which is the object, and the AUROC collapses. This is precisely the generative assumption—static scene, rigid camera, moving objects as a minority—that a localization theorem for H^1 would need to state, now made empirically concrete, with the static fraction as the natural control parameter for a separation bound.

The linearization is valid in pose error, not object motion. Because object motion enters only through the (exactly computed) residual r^0, while the linearization is in the *pose corrections* ξ, the familiar "large motion breaks the tangent approximation" concern attaches to the magnitude of pose error, not to how fast the object moves. Empirically the harmonic localization is invariant to camera drift up to large per-step magnitudes and degrades only at extreme drift, far outside any realistic regime; the valid operating window is wide. Conversely, the projection removes only pose-explainable disagreement, so it offers no protection against per-pixel measurement noise—both readings fail once noise rivals the object's motion.

3.4 Scope

This is a synthetic, in-the-clean test: the residuals are computed from ground-truth geometry rather than from a learned predictor, which is what lets it isolate the sheaf machinery from prediction quality. It establishes that *if* the pairwise predictions are clean enough, the harmonic H^1 provably localizes scene dynamics and does so with an invariance to pose drift that the raw residual lacks. It does not yet establish that a real pointmap network's residuals meet that bar, nor does it treat occlusion—which is missing data rather than inconsistency and warrants its own test—and the static-dominance precondition remains a conjecture awaiting the separation bound. Validating the construction on D^2USt3R predictions for a real dynamic clip is the immediate next step.

References

J. M. Curry. *Sheaves, cosheaves and applications*. University of Pennsylvania, 2014.

R. W. Ghrist. *Elementary applied topology*, volume 1. Createspace Seattle, 2014.

J. Han, H. An, J. Jung, T. Narihira, J. Seo, K. Fukuda, C. Kim, S. Hong, Y. Mitsufuji, and S. Kim. Enhancing 3d reconstruction for dynamic scenes. *Advances in Neural Information Processing Systems*, 38:1210–1234, 2026.

J. Hansen and R. Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3(4):315–358, 2019.

J. Hansen and R. Ghrist. Opinion dynamics on discourse sheaves. *SIAM Journal on Applied Mathematics*, 81(5): 2033–2060, 2021.

V. Leroy, Y. Cabon, and J. Revaud. Grounding image matching in 3d with mast3r. In *European conference on computer vision*, pages 71–91. Springer, 2024.

R. Murai, E. Dexheimer, and A. J. Davison. Mast3r-slam: Real-time dense slam with 3d reconstruction priors. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 16695–16705, 2025.

M. Robinson. Sheaves are the canonical data structure for sensor integration. *Information Fusion*, 36:208–224, 2017.

D. M. Rosen, L. Carlone, A. S. Bandeira, and J. J. Leonard. Se-sync: A certifiably correct algorithm for synchronization over the special euclidean group. *The International Journal of Robotics Research*, 38(2-3):95–125, 2019.

A. Singer. Angular synchronization by eigenvectors and semidefinite programming. *Applied and computational harmonic analysis*, 30(1):20–36, 2011.

H. Wang and L. Agapito. 3d reconstruction with spatial memory. In *2025 International Conference on 3D Vision (3DV)*, pages 78–89. IEEE, 2025.

Q. Wang, Y. Zhang, A. Holynski, A. A. Efros, and A. Kanazawa. Continuous 3d perception model with persistent state. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 10510–10522, 2025.

S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20697–20709, 2024.

X. Yu and H. Yang. Sim-sync: from certifiably optimal synchronization over the 3d similarity group to scene reconstruction with learned depth. *IEEE Robotics and Automation Letters*, 9(5):4471–4478, 2024.

J. Zhang, C. Herrmann, J. Hur, V. Jampani, F. Cole, D. Sun, M.-H. Yang, et al. Monst3r: A simple approach for estimating geometry in the presence of motion. In *International Conference on Learning Representations*, volume 2025, pages 82863–82886, 2025.